Name and Surname:	
Subject:	Mathematics
Grade:	11AG
Duration:	40 minutes
Exam:	Final Exam
Name of the Teacher:	Anvarbek Khaydarov
Date:	26/10/2026
Score:	_____/100%

INSTRUCTIONS

1. Read all questions carefully before answering.
2. Use a blue pen or pencil to answer all questions.
3. Write your answer to each question in the space provided.
4. No electronic devices (phones, smartwatches, headphones) are allowed.
5. Do not leave the class/exam hall in the first 30 minutes

Section A — Easy (6 questions × 4 marks = 24 marks)
Answer all six questions.

1. Find $\lim_{x \to 2} (3x^2 - 5)$. (4 marks)

2. Find $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$. (4 marks)

3. Find $\lim_{x \to \infty} \dfrac{5x^2 + 1}{2x^2 - x}$. (4 marks)

4. Differentiate $y = 4x^5$. (4 marks)

5. Differentiate $y = \sin x + e^x$. (4 marks)

6. Differentiate $y = \ln x - \cos x$. (4 marks)

Section B — Medium (6 questions × 6 marks = 36 marks)
Answer all six questions. Show your working.

7. Find $\lim_{x \to 1} \dfrac{x^2 + 2x - 3}{x - 1}$. (6 marks)

8. Find $\lim_{x \to \infty} \dfrac{3x^3 - 2x}{6x^3 + 5}$. (6 marks)

9. Differentiate $y = (2x + 1)^5$. (6 marks)

10. Differentiate $y = x^2 e^x$, giving your answer in factorised form. (6 marks)

11. Differentiate $y = \tan 4x$, and find the gradient of the curve at $x = 0$. (6 marks)

12. Differentiate $y = \ln\left(3x^2 + 1\right)$. (6 marks)

Section C — Hard (3 questions, 40 marks)
Answer all three questions. Show every step of your working.

13. Evaluate the following limits: (13 marks)

(a) $\displaystyle\lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$ (6 marks)

(b) $\displaystyle\lim_{x \to \infty} \left(\sqrt{x^2 + 3x} - x \right)$ (7 marks)

14. Differentiate, simplifying your answer where possible: (13 marks)

(a) $y = \sqrt{x^2 + 9}$ (6 marks)

(b) $y = \ln(\cos x)$ (7 marks)

15. A curve has equation $y = x^2 \sin x$. (14 marks)

(a) Find $\dfrac{dy}{dx}$. (7 marks)

(b) Find $\dfrac{d^2y}{dx^2}$. (7 marks)

— End of the exam —